

Óscar A. Chávez Ortiz

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Summary

Data Scientist and Ph.D. researcher with 8+ years of experience in large-scale data preprocessing, statistical modeling, and machine learning. Proven expertise in building end-to-end quantitative pipelines for ETL, model development, optimization, and uncertainty quantification using frequentist and Bayesian methods. Extensive experience with Bayesian inference, MCMC, simulation-based inference, and high-performance computing. Published in peer-reviewed journals and skilled at communicating complex technical results to both technical and non-technical audiences.

Technical Skills

Languages:	Python, SQL, Git/GitHub, R,
ML Frameworks:	scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, sbi
ML Techniques:	Regression, Classification, Neural Networks (NN), Convolutional Neural Networks (CNN), Random Forest, XGBoost, Simulation-Based Inference
Statistical Methods:	Model Training and Evaluation, MCMC, Uncertainty Quantification
Visualization:	Matplotlib, Seaborn, Tableau
Data Tools:	Pandas, NumPy, SciPy, Astropy, h5py, Photutils, emcee
Communication:	Presentations, Scientific Writing, Media Commentary, Outreach Talks

Work Experience

Graduate Student Researcher

August 2020 – Present

The University of Texas at Austin

Project: Bayesian Model Selection and Parameter Estimation

January 2025 – November 2025

Goal: Discover combination of physical processes best explains the observed features of a very distant galaxy.

- Develop Bayesian model comparisons to select the most likely model from a suite of models, using the Bayesian evidence to guide inference and comparison.
- Executed large-scale parameter estimation to infer key system properties, including uncertainty quantification via Bayesian modeling.
- Developed workflows for model evaluation, enabling systematic benchmarking and comparison across a complex parameter grid.
- Write results to inform theoretical scenarios for early galaxy and black hole formation, connecting model outputs to physically meaningful conclusions in a peer-reviewed Journal.

Project: End-to-End Pipeline Performance Evaluations

September 2025 – October 2025

Goal: Assess and correct for biases in data processing pipeline to ensure accurate measurements from datasets.

- Developed a custom automated synthetic data generation script and framework to measure pipeline completeness and detect failure modes.
- Diagnosed and resolved pipeline issues through systematic testing, improving total data processing reliability.
- Applied completeness corrections to imaging and spectral data, ensuring that the final results were physically meaningful and unbiased for downstream modeling.

Project: Bayesian Inference & Predictive Modeling Pipeline

August 2023 – December 2024

Goal: Develop a Bayesian inference pipeline to extract physical properties from complex multi-dimensional datasets, quantify uncertainties, and identify key patterns in observational data.

- Designed and developed a new Bayesian inference pipeline to analyze large-scale observational datasets into a single unified algorithm to uncover key drivers of signal detectability.
- Automated ETL of multiple data sources, performed data quality checks and custom analysis workflows to flag good quality data.

- Built custom statistical models and ran Bayesian fitting with *emcee* to robustly measure system parameters and accurately quantify their uncertainty. The work resulted in a peer-reviewed journal scientific journal.

Project Pilot Study: Feature-Observable Correlation Analysis

August 2020 – August 2023

Goal: Explore meaningful correlations between features and target observables from a minimal dataset, establishing a proof-of-concept for scalable ML pipelines.

- Led and developed a custom end-to-end data analysis pipeline, including data ingestion, cleaning, and feature selection, to identify high-quality samples for further modeling.
- Performed ETL on multiple datasets to be able to apply statistical techniques (Spearman correlation, and bootstrapping) to quantify relationships and associated uncertainties, being the first to find evidence for correlations to system observability with this dataset.
- The findings of this work were presented in many presentations to both technical and non-technical audiences.
- This result also led to a peer-reviewed paper in a well respected scientific journal.

Other Work Experience

Summer Internship

May 2025 - August 2025

The University of Texas at Austin

Project: Auditory Health Care Analysis

Goal: Using ML/AI Models to determine human pain level.

- Developed a custom deep learning pipeline to classify raw .wav audio files as indicative of pain or not, converting audio to spectrograms for model input.
- Developed a neural network from scratch, then leveraged a pre-trained Hugging Face model (CLAP) to improve auditory classification performance using transfer learning.
- Evaluated model generalization via randomized and patient-level splits, identifying a significant performance drop (75% → 60%) when testing on unseen patients, highlighting the importance of robust evaluation strategies and careful care in the training/testing set.

Team Leader

January 2025 – May 2025

The University of Texas at Austin

Project: Bayesian Model Selection and Parameter Estimation

Goal: Determine which machine learning model performs the best at predicting an observable from a dataset.

- Led a team of four to compare multiple ML models (Neural Networks, XGBoost, Linear Regression, Random Forest) on a shared dataset, coordinating methodology and ensuring consistent evaluation metrics.
- Developed a custom script that performed ETL, data cleaning, model building and training using PyTorch.
- Consolidated results and designed a unified evaluation framework to benchmark model performance across the different methods.
- Mentored team members in statistical modeling, ML evaluation, and reproducible coding practices, fostering collaboration and accelerating project progress.
- Presented this work and wrote a report that explained the workflow, methods, and results from the model comparison.

Education

Ph.D in Astronomy

Expected Spring 2026

The University of Texas at Austin

M.S. in Astronomy

Spring 2024

The University of Texas at Austin

B.A. in Astrophysics

December 2019

University of California, Berkeley